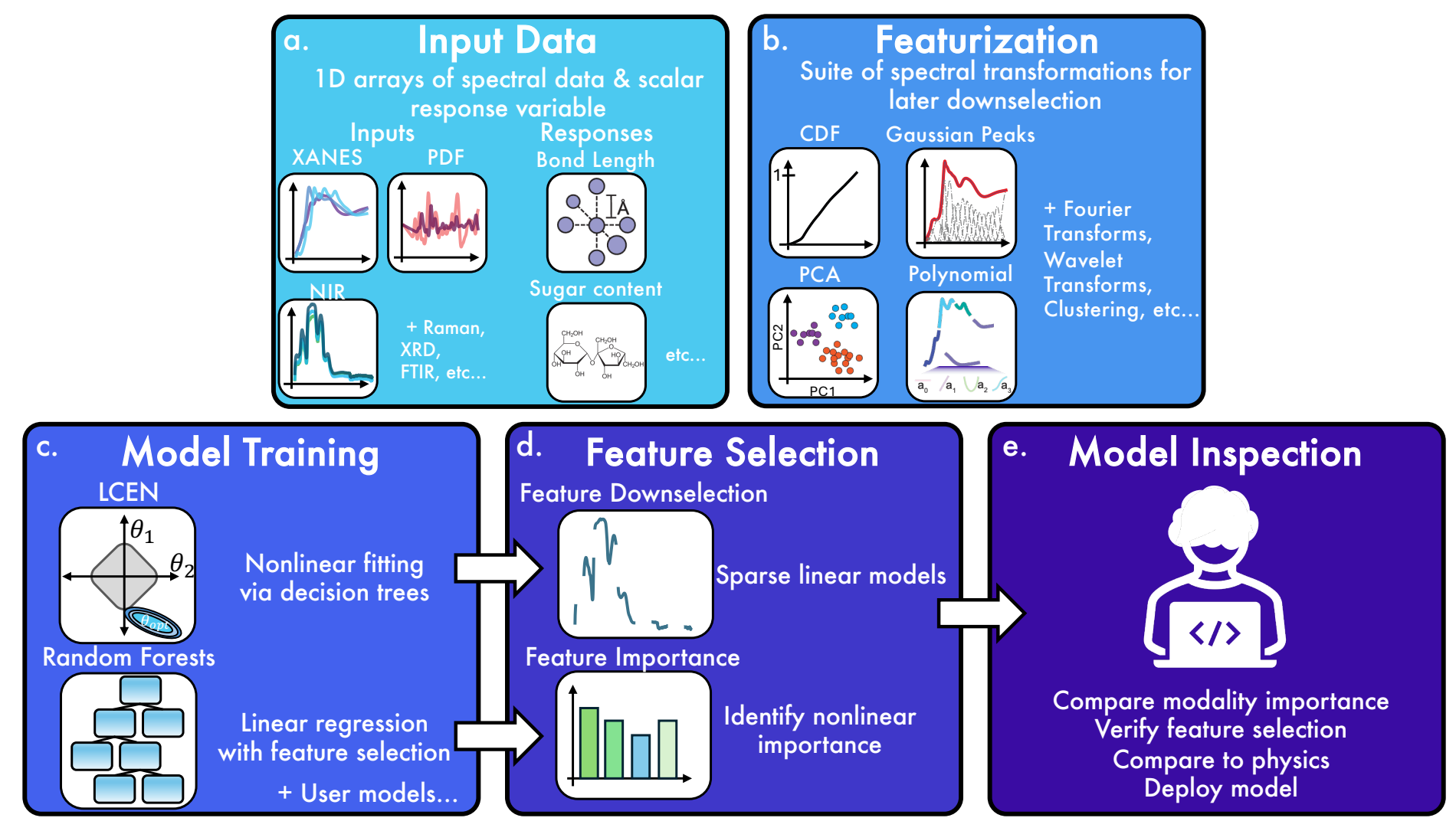

a.
Input Data
1D arrays of spectral data & scalar response variable
Inputs
Responses
XANES
PDF
Bond Length
NIR
Sugar content
+ Raman, XRD, FTIR, etc...
etc...
b.
Featurization
Suite of spectral transformations for later downselection
CDF
Gaussian Peaks
PCA
Polynomial
PC2
PC1
+ Fourier Transforms, Wavelet Transforms, Clustering, etc...
c.
Model Training
LCEN
Random Forests
Nonlinear fitting via decision trees
Linear regression with feature selection
+ User models...
d.
Feature Selection
Feature Downselection
Sparse linear models
Feature Importance
Identify nonlinear importance
e.
Model Inspection
Compare modality importance
Verify feature selection
Compare to physics
Deploy model